

Problem solving and testing using java

WEEK 1 TASKS

20/7/2026-25/7/2026

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Task 1: Access and print the element at a given index in an array.

Task 2: Search for a given element in a sorted array using Binary Search.

Task 3: Find the maximum element in an array of n integers.

Task 4: Find the Kth smallest element in an array.

Task 5: Print all possible pairs of elements from an array.

Task 6: Sum of even or odd digits.

Task 7: Nth Fibonacci.

Task 8: Check whether a number is a Palindrome.

Task 9: Sum of the last digits of two given numbers.

TASK 1

```
package lambda_function; import  
java.util.Scanner;
```

```
@FunctionalInterface    interface
```

```
ArrayElement {    void display(int[]
```

```
arr, int index);
```

```
}
```

```
public class TASK1 {    public static void
```

```
main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    System.out.print("Enter  array  size: ");
```

```
    int n = sc.nextInt();
```

```
    int arr[] = new int[n];
```

```
    System.out.println("Enter array elements:");
```

```
    for (int i = 0; i < n; i++) {
```

```
arr[i] = sc.nextInt();
```

```
    }
```

```
    System.out.print("Enter  index: ");
```

```
    int index = sc.nextInt();
```

```
    //    Lambda    Expression
```

```
    ArrayElement obj = (a, i) -> {
```

```
    if (i >= 0 && i < a.length)
```

```
        System.out.println("Element  =  " + a[i]);
```

```
    else
```

```

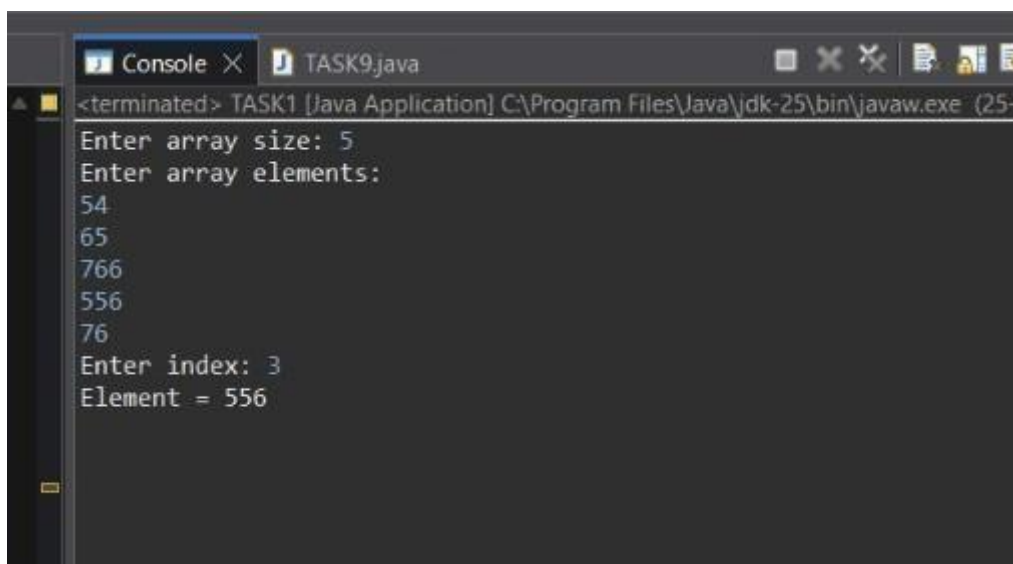
        System.out.println("Invalid Index, that index is not there");
    };

    obj.display(arr, index);

//    sc.close();
}
}

```

OUTPUT:



```

<terminated> TASK1 [Java Application] C:\Program Files\Java\jdk-25\bin\javaw.exe (25-)
Enter array size: 5
Enter array elements:
54
65
766
556
76
Enter index: 3
Element = 556

```

```

TASK          2          package
lambda_function;      import
java.util.Scanner;

@FunctionalInterface interface
BinarySearch {          void
search(int[] arr, int key);

```

```

}

public class TASK2 {    public static void
main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter array size: ");    int
n = sc.nextInt();    int arr[] = new int[n];

    System.out.println("Enter sorted array elements:");

    for (int i = 0; i < n; i++) {
arr[i] = sc.nextInt();
    }

    System.out.print("Enter element to search: ");

    int key = sc.nextInt();

    // Lambda Expression
    BinarySearch bs = (a, k) -> {
int low = 0;        int high =
a.length - 1;        boolean
found = false;        while (low
<= high) {        int mid = (low
+ high) / 2;

        if (a[mid] == k) {

            System.out.println("Element found at index " + mid);

            found = true;        break;

        } else if (a[mid] < k) {

            low = mid + 1;

        }        else        {

            high = mid - 1;

        }

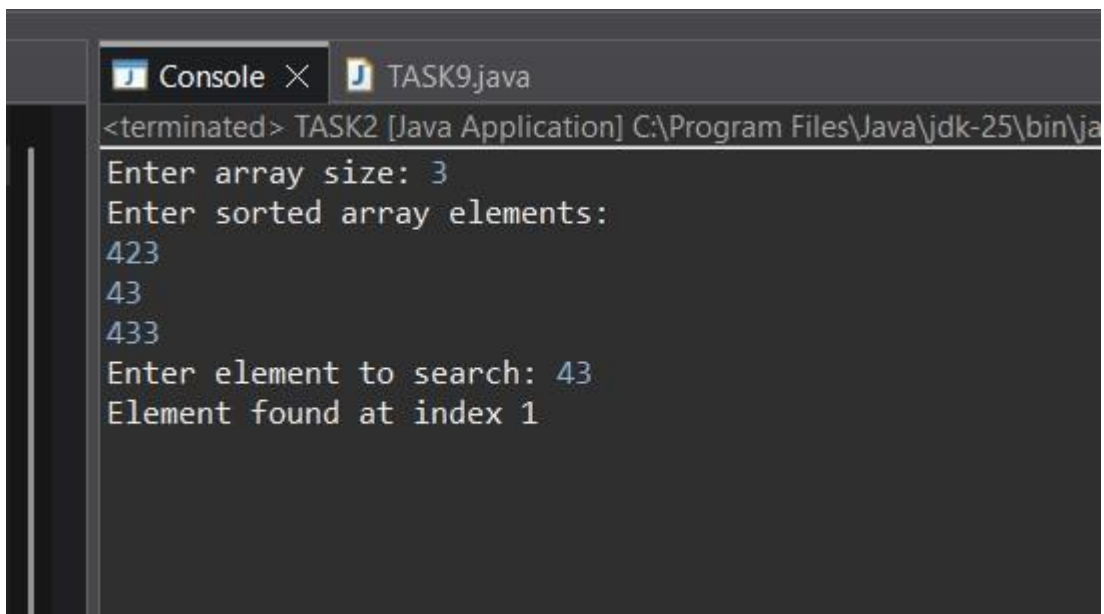
```

```

    }
    if (!found) {
        System.out.println("Element not found");
    }
};
bs.search(arr, key);
sc.close();
}
}

```

OUTPUT:



```

Console x TASK9.java
<terminated> TASK2 [Java Application] C:\Program Files\Java\jdk-25\bin\ja
Enter array size: 3
Enter sorted array elements:
423
43
433
Enter element to search: 43
Element found at index 1

```

```

TASK      3      package
lambda_function;  import
java.util.Scanner;
@FunctionalInterface
interface Maximum {
    void findMax(int[] arr);
}

```

```

public class TASK3 {    public static void
main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter array size: ");    int
n = sc.nextInt();    int arr[] = new int[n];

    System.out.println("Enter array elements:");

    for (int i = 0; i < n; i++) {
arr[i] = sc.nextInt();
    }

    //    Lambda    Expression
Maximum maxElement = (a) -> {
    int max = a[0];    for (int i
= 1; i < a.length; i++) {    if
(a[i] > max) {    max = a[i];
    }
    }

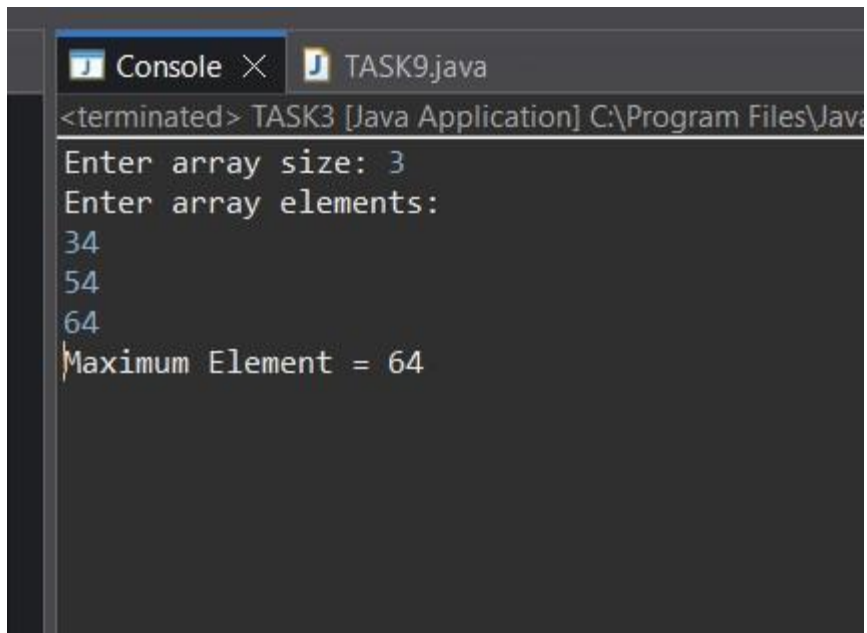
    System.out.println("Maximum Element = " + max);    };

    maxElement.findMax(arr);

    sc.close();
    }
}

```

OUTPUT:



```
<terminated> TASK3 [Java Application] C:\Program Files\Java
Enter array size: 3
Enter array elements:
34
54
64
Maximum Element = 64
```

```
TASK 4 package lambda_function;
import java.util.Arrays; import
java.util.Scanner;

@FunctionalInterface interface
KthSmallestElement { void
findKthSmallest(int[] arr, int k);
}
```



```

public class TASK4 {    public static void
main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter  array  size:  ");
int n = sc.nextInt();

    int arr[] = new int[n];

    System.out.println("Enter array elements:");
    for (int i = 0; i < n; i++) {
arr[i] = sc.nextInt();
    }

    System.out.print("Enter  K:  ");
int k = sc.nextInt();

    // Lambda Expression
    KthSmallestElement obj = (a, key) -> {        Arrays.sort(a);

        if (key >= 1 && key <= a.length) {
            System.out.println("Kth Smallest Element = " + a[key - 1]);
        } else {
            System.out.println("Invalid K");
        }
    };
};

```

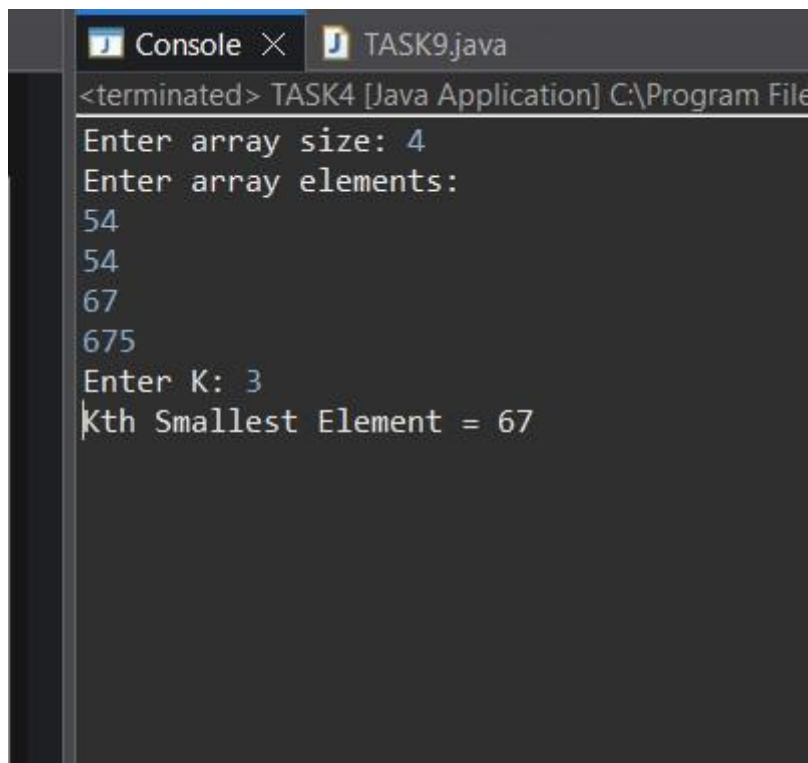
```

        obj.findKthSmallest(arr, k);

        sc.close();
    }
}

```

OUTPUT :



```

<terminated> TASK4 [Java Application] C:\Program File
Enter array size: 4
Enter array elements:
54
54
67
675
Enter K: 3
Kth Smallest Element = 67

```

```

TASK      5      package
lambda_function;    import
java.util.Scanner;

@FunctionalInterface

interface ArrayPair {    void
printPairs(int[] arr);
}

public class TASK5 {

```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter array size: ");
    int n = sc.nextInt();

    int arr[] = new int[n];

    System.out.println("Enter array elements:");
    for (int i = 0; i < n; i++) {
        arr[i] = sc.nextInt();
    }

    //      Lambda      Expression
    ArrayPair pair = (a) -> {
        System.out.println("Pairs:");

        for (int i = 0; i < a.length; i++) {
            for (int j = i + 1; j < a.length; j++) {
                System.out.println(a[i] + " " + a[j]);
            }
        }
    };

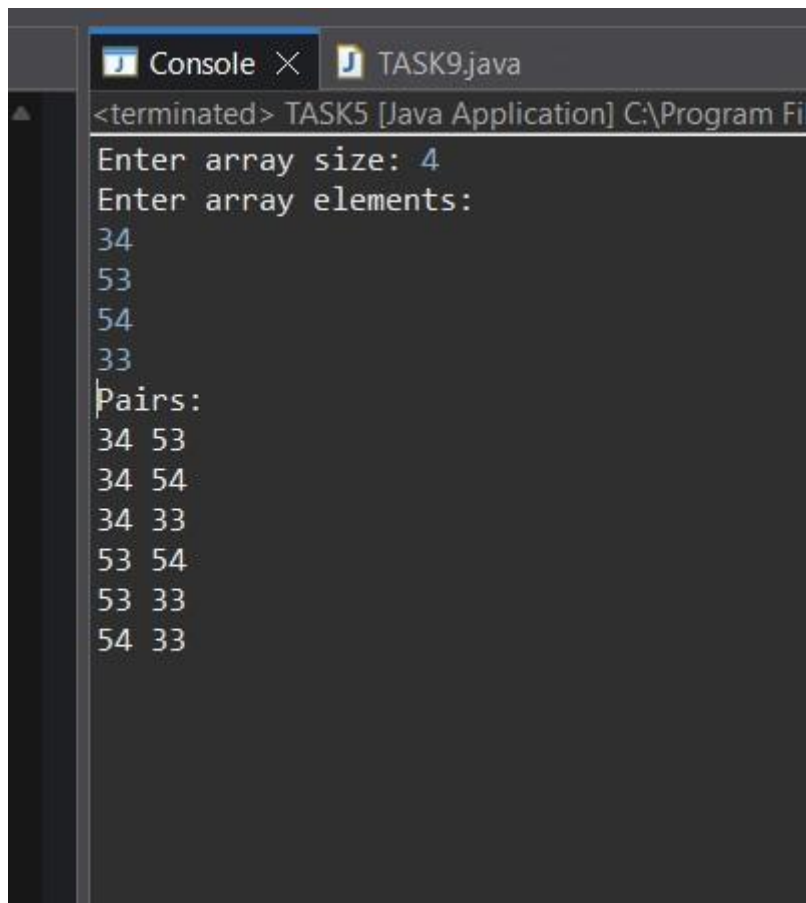
    pair.printPairs(arr);

    sc.close();
}

```

```
}  
}
```

OUTPUT :



```
<terminated> TASK5 [Java Application] C:\Program Fi  
Enter array size: 4  
Enter array elements:  
34  
53  
54  
33  
pairs:  
34 53  
34 54  
34 33  
53 54  
53 33  
54 33
```

```
package lambda_function;
```

TASK 6

```
import java.util.Scanner;
```

```
@FunctionalInterface interface DigitSum {
```

```
void findSum(int num, String choice);
```

```
}
```

```
public class TASK6 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");        int
```

```
num = sc.nextInt();
```

```
        System.out.print("Enter choice (even/odd): ");
```

```
        String choice = sc.next();
```

```
        //        Lambda        Expression
```

```
        DigitSum ds = (number, ch) -> {
```

```
int sum = 0;        while (number > 0)
```

```
{            int digit = number % 10;
```

```
                if (ch.equalsIgnoreCase("even") && digit % 2 == 0) {
```

```
sum += digit;
```

```
                }
```

```
                if (ch.equalsIgnoreCase("odd") && digit % 2 != 0) {
```

```
sum += digit;
```

```
                }
```

```
                number /= 10;
```

```
}
```

```
System.out.println("Sum = " + sum);
```

```
};
```

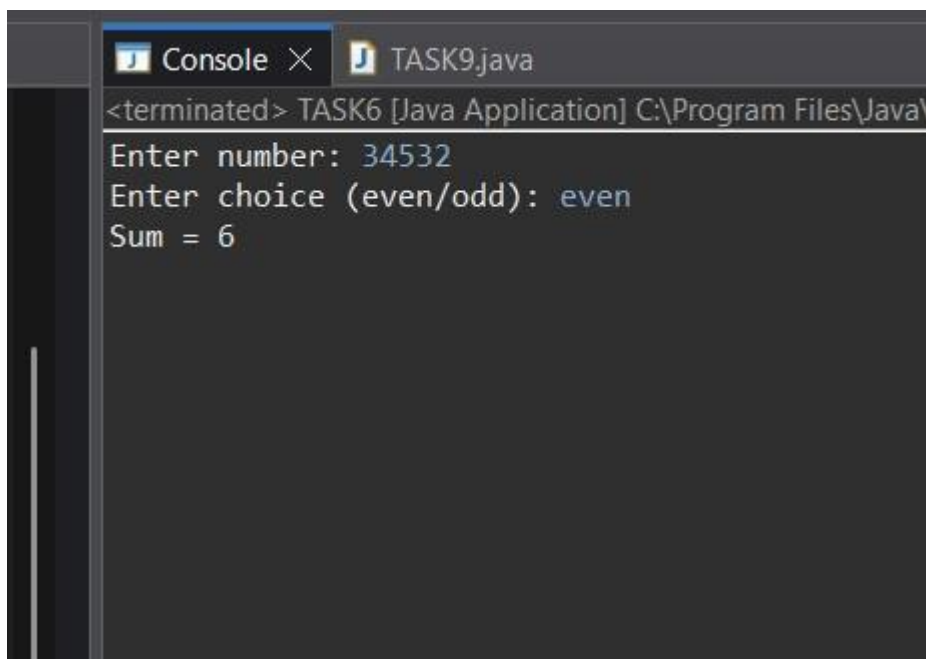
```
ds.findSum(num, choice);
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT



```
Console x TASK9.java
<terminated> TASK6 [Java Application] C:\Program Files\Java\
Enter number: 34532
Enter choice (even/odd): even
Sum = 6
```

TASK 7

```
package lambda_function;
```

```
import java.util.Scanner;
```

```
@FunctionalInterface interface
```

```
FibonacciNumber {    void
```

```
findFibonacci(int n);
```

```
}
```

```
public class TASK7 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter N: ");
```

```
        int n = sc.nextInt();
```

```
        // Lambda Expression
```

```
        FibonacciNumber fib = (num) -> {
```

```
            int a = 0, b = 1;
```

```
            if (num == 1) {
```

```
                System.out.println("Nth Fibonacci = " + a);
```

```
            }
```

```
            else if (num == 2) {
```

```
                System.out.println("Nth Fibonacci = " + b);
```

```
            }
```

```

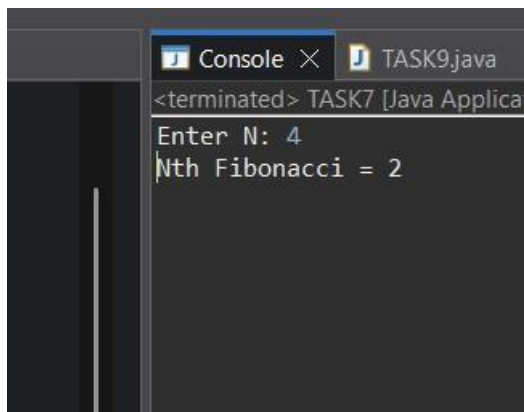
        else {
int c = 0;

        for (int i = 3; i <= num; i++) {
c = a + b;          a = b;
b = c;
        }

        System.out.println("Nth Fibonacci = " + c);
    }
};
fib.findFibonacci(n);
sc.close();
}
}

```

OUTPUT:



The screenshot shows a Java IDE with two tabs: 'Console' and 'TASK9.java'. The 'Console' tab is active and displays the following text: '<terminated> TASK7 [Java Applica', 'Enter N: 4', and 'Nth Fibonacci = 2'. The text is white on a dark background.

TASK 8

```
import java.util.Scanner;
```



```
package lambda_function;
```

```
@FunctionalInterface interface
```

```
Palindrome { void
```

```
checkPalindrome(int num);
```

```
}
```

```
public class TASK8 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter number: ");
```

```
        int num = sc.nextInt();
```

```
        // Lambda Expression
```

```
        Palindrome p = (number) -> {
```

```
            int original = number;
```

```
            int reverse = 0;
```

```
            while (number > 0) {                int
```

```
                digit = number % 10;            reverse
```

```
                = reverse * 10 + digit;        number
```

```
                /= 10;
```

```
            }
```

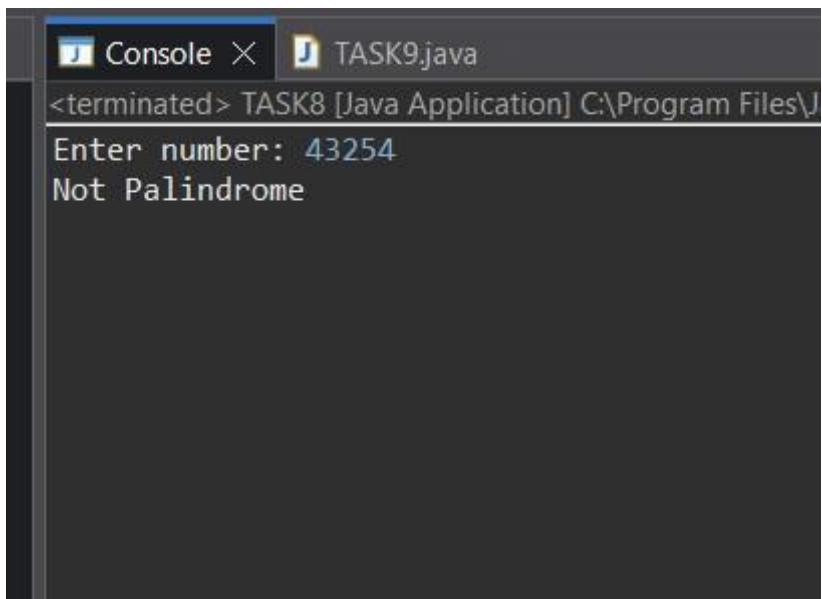
```
            if (original == reverse)
```

```
        System.out.println("Palindrome");
else
        System.out.println("Not Palindrome");
};

p.checkPalindrome(num);

sc.close();
}
}
```

OUTPUT



The screenshot shows a Java IDE window with two tabs: 'Console' and 'TASK9.java'. The 'Console' tab is active, displaying the following text: '<terminated> TASK8 [Java Application] C:\Program Files\J'. Below this, the program's output is shown: 'Enter number: 43254' followed by 'Not Palindrome' on the next line. The input '43254' is highlighted in blue, indicating it was entered by the user.

TASK 9 package

lambda_function;

import java.util.Scanner;

@FunctionalInterface interface

LastDigitSum { void

sumLastDigits(int a, int b);

}

public class TASK9 {

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

System.out.print("Enter first number: ");

int a = sc.nextInt();

System.out.print("Enter second number: ");

int b = sc.nextInt();

// Lambda Expression

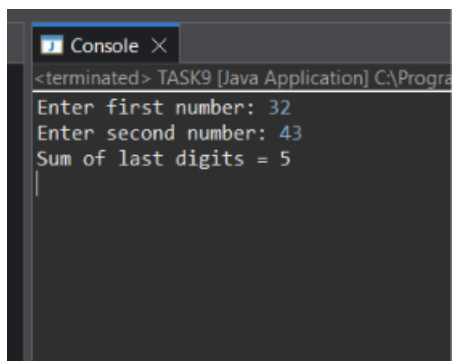
LastDigitSum sum = (x, y) -> {

int last1 = x % 10;

int last2 = y % 10;

```
System.out.println("Sum of  
last digits = " + (last1 +  
last2));  
  
};  
  
sum.sumLastDigits(a, b);  
  
sc.close();  
}  
}
```

OUTPUT



```
Console X  
<terminated> TASK9 [Java Application] C:\Progra  
Enter first number: 32  
Enter second number: 43  
Sum of last digits = 5  
|
```